

Development and validation of prediction algorithm to identify tuberculosis in two large California health systems

Corresponding Author: Dr Heidi Fischer

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

Congratulations to the author on this analysis. This is a great use of a large health provider database to solve an important problem of how to meaningfully screen the right patients for LTBI in order to prevent TB morbidity. The use of the KPSC cohort for your training set and the external validation with the KPNC cohort elegantly provides similar cohorts, especially in regard to access to health insurance and access.

I have made the following suggestions to clarify your message to, please accept if useful:

line 49: Suggest clarifying "largest" in "largest number of people with TB disease in the 49 United States (US), with 22% of cases". Do you mean highest absolute number or largest proportion in the U.S.

line 57: Suggest adding "LTBI" before screening and treatment efforts to delineate from TB screening and treatment

line 62: suggest edit: "close contact to SOMEONE WITH TB disease"

line 66: suggest using "during" instead of "in"

line 69/70: The meaning of this run-on sentence was not clear to me on multiple readings. I would suggest you re-writing and perhaps breaking into multiple sentences for clearer communication.

line 78: suggest replacing "in up to" with "for"

Lastly, I found the supplementary tables, particularly supplementary table S2, very interesting and helpful for any other health systems that might want to do something similar. If allowed, you may consider including that in the main tables, or substituting something else out.

Thanks for the opportunity to review this interesting and important work.

(Remarks on code availability)

Reviewer #2

(Remarks to the Author)

Thank you to the authors and editors for the opportunity to review this interesting manuscript. Fischer et al use EHR datasets from California to develop and validate a multivariable prognostic model for incident tuberculosis. The intention is for the model to direct latent TB testing and treatment efforts towards those at highest risk. The authors find that using their model to prioritise screening efforts could reduce the number needed to screen, where resource constraints limit the overall number of people who can be tested for LTBI. Where widespread testing of at risk populations is possible, the model does not demonstrate added value.

Overall, the analyses are reasonably well conducted and transparently reported. The effort to target LTBI testing is perfectly justifiable given that absolute risks of disease are very low among most populations, with many services historically seeking to implement testing too broadly in low-incidence countries. I have itemised my comments as major and minor below.

Major

1. Generalisability beyond California is a potential concern, since only Californian data were used in development and validation. Do the authors see this as a tool for local use only, or do they envisage wider evaluation and implementation? If the latter, have they considered seeking external validation for example in Canada, where similar datasets exist (<https://academic.oup.com/cid/advance-article/doi/10.1093/cid/ciae561/7904918>)?
2. As a related point, it would be useful to know how well the Canadian model (which is intended for similar use) performs in the Californian data, as a benchmarking exercise (<https://academic.oup.com/cid/advance-article/doi/10.1093/cid/ciae561/7904918>)?
3. The lack of calibration assessment is also a major concern. It is highly unusual not to report calibration performance, even for a rare outcome. This is certainly strongly recommended in TRIPOD reporting guidance and I would urge the authors to assess this formally.

Minor

1. The abstract refers to the 'simulated' TB outcome, but not the observed TB outcome which I believe was also assessed?
2. Lines 98-99 - is this the actual numbers who tested positive for LTBI during the study period? Suggest further clarification of this
3. Line 104 - Is it worth clarifying that linear and spline forms are only relevant to age and BMI? Or are there other variables here that I have missed?
4. Consider moving Table S4 (model coefficients) to the main manuscript, since this feels at the "heart" of the paper

(Remarks on code availability)

I did not see a link to the code. Apologies if I missed this

Reviewer #3

(Remarks to the Author)

I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

(Remarks on code availability)

Version 1:

Reviewer comments:

Reviewer #2

(Remarks to the Author)

Many thanks to the authors for their responses. They have done a thorough job of addressing the reviewers' comments. I have 2 further recommendations:

1. Given that the authors acknowledge that this is currently a California-specific tool (given the absence of validation beyond the state), please consider adding "in California" as 2 additional words at the end of the final sentence of the abstract.
2. The calibration assessment could still be improved. The authors present "mean calibration". By this, do they mean calibration in the large (<https://www.bmj.com/content/386/bmj-2023-078276>)? If so, the usual ideal target value would be 0, rather than 1. This requires further explanation in the methods section, including how it was calculated and a citation. Alternatively, they could present observed:expected ratios instead of mean calibration. I also recommend adding calibration plots.

(Remarks on code availability)

Reviewer #3

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(Remarks on code availability)

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REVIEWER COMMENTS

We thank the reviewers for their helpful comments and feel the manuscript has been much improved with these revisions.

Reviewer #1 (Remarks to the Author):

Congratulations to the author on this analysis. This is a great use of a large health provider database to solve an important problem of how to meaningfully screen the right patients for LTBI in order to prevent TB morbidity. The use of the KPSC cohort for your training set and the external validation with the KPNC cohort elegantly provides similar cohorts, especially in regard to access to health insurance and access.

I have made the following suggestions to clarify your message to, please accept if useful:

line 49: Suggest clarifying "largest" in "largest number of people with TB disease in the 49 United States (US), with 22% of cases". Do you mean highest absolute number or largest proportion in the U.S.

Thank you for pointing this out. We meant the highest absolute number and have added this clarification to the text.

line 57: Suggest adding "LTBI" before screening and treatment efforts to delineate from TB screening and treatment

This is an excellent point and we have added this to the text.

line 62: suggest edit: "close contact to SOMEONE WITH TB disease"

We have added this to the text.

line 66: suggest using "during" instead of "in"

This does read more clearly, and we have changed the wording as suggested.

line 69/70: The meaning of this run-on sentence was not clear to me on multiple readings. I would suggest you re-writing and perhaps breaking into multiple sentences for clearer communication.

We agree this sentence needs to be clarified and have edited and broke it down to multiple sentences. We hope it is improved.

line 78: suggest replacing "in up to" with "for"

We have edited this text and hope it is clearer.

Lastly, I found the supplementary tables, particularly supplementary table S2, very interesting and helpful for any other health systems that might want to do something similar. If allowed, you may consider including that in the main tables, or substituting something else out.

We agree with the reviewer's suggestion and former supplementary table S2 has been moved from supplementary materials to main Table 2.

Thanks for the opportunity to review this interesting and important work.

Reviewer #2 (Remarks to the Author):

Thank you to the authors and editors for the opportunity to review this interesting manuscript. Fischer et al use EHR datasets from California to develop and validate a multivariable prognostic model for incident tuberculosis. The intention is for the model to direct latent TB testing and treatment efforts towards those at highest risk. The authors find that using their model to prioritise screening efforts could reduce the number needed to screen, where resource constraints limit the overall number of people who can be tested for LTBI. Where widespread testing of at risk populations is possible, the model does not demonstrate added value.

Overall, the analyses are reasonably well conducted and transparently reported. The effort to target LTBI testing is perfectly justifiable given that absolute risks of disease are very low among most populations, with many services historically seeking to implement testing too broadly in low-incidence countries. I have itemised my comments as major and minor below.

Major

1. Generalisability beyond California is a potential concern, since only Californian data were used in development and validation. Do the authors see this as a tool for local use only, or do they envisage wider evaluation and implementation? If the latter, have they

considered seeking external validation for example in Canada, where similar datasets exist (<https://academic.oup.com/cid/advance-article/doi/10.1093/cid/ciae561/7904918>)?

This model was developed and validated using data from California, and thus at this time it is intended for use in health systems or by health providers within the state. Given California's large population and burden of TB disease, this tool can already provide great value. However, as pointed out by the reviewer, it could also be useful as a TB disease prediction tool outside of California, and validation here is an important direction for future research.

We thank the reviewer for bringing to our attention the TB disease prediction algorithm developed in Canada, a model that will be very useful in improving directed LTBI screening. The Canadian model referenced by the reviewer has similar goals for focused, efficient screening of TB disease in countries with lower TB disease incidence; however, the target populations in the Canadian study versus our study are different, prohibiting direct comparisons between the two models. Specifically, the Canadian model aims to predict TB disease for persons recently (within 5 years) immigrating to Canada. In comparison, our model predicts TB disease among the full patient populations in two large health systems in California. While all patients used to develop and validate the Canadian model will be foreign-born and have arrived in Canada no more than 5 years prior to baseline, most patients in our population are born in the United States, and a large percentage of our foreign-born patients have lived in the United States for more than the 5-year time frame of the Canadian model. Foreign-born status is used as a predictor in our model, and there would be no variation for this predictor using the Canadian data.

Whereas the Canadian prediction model fills the important role of targeting TB disease for foreign-born individuals recently entering a low TB incidence country, the goal of our model is to better target LTBI screening among the larger population living in low TB incidence countries, including foreign-born patients immigrating to the US more than 5 years ago as well as other higher risk populations, such as immunosuppressed persons. In recent years, more TB disease diagnoses among foreign born individuals in the U.S. have occurred in the 10 years after arrival in the U.S. than among those in the U.S. <10 years (citation provided in manuscript revision). As such, effective LTBI screening requires moving beyond the public health sector screening of recent immigrants to the U.S., to expand partnerships with health care providers. This new risk prediction algorithm is a valuable tool for improving the efficiency of LTBI screening done by health care providers, who are working with information

available in their EHRs, and do not have access to information supplied to immigration authorities. We believe the Canadian model and the model developed in this manuscript can work in tandem with the common goal of decreasing TB disease incidence overall.

We have added information comparing our model to the Canadian model in the Discussion, providing more context to how each can be a useful tool to decrease overall TB disease burden and make LTBI screening more efficient in lower TB incidence countries. Additionally, we have clarified how we currently envision our model's use in California, and that validation in other geographic regions is an important direction for future research. We believe these additions have strengthened this manuscript and thank the reviewer again for these helpful insights.

2. As a related point, it would be useful to know how well the Canadian model (which is intended for similar use) performs in the Californian data, as a benchmarking exercise (<https://academic.oup.com/cid/advance-article/doi/10.1093/cid/ciae561/7904918>)?

As outlined in our previous response, this unfortunately cannot be accomplished because the Canadian model is meant to be used specifically for recent immigrants, while the California data includes the general population of two large healthcare systems. Importantly, the Canadian model requires inputs on the length of time the patient has lived in the United States as well as refugee status, which is not information available in our EHR. Even if it was available, it would not be applicable to the majority of patients in the populations of the California data who were born in the United States or have been living in the United States for more than 5 years. We believe, however, that the additional text added to the Discussion section comparing the Canadian model to the model developed in this paper provide further context to reader as to how these tools are meant to be used to improve LTBI screening in lower incidence TB disease countries.

3. The lack of calibration assessment is also a major concern. It is highly unusual not to report calibration performance, even for a rare outcome. This is certainly strongly recommended in TRIPOD reporting guidance and I would urge the authors to assess this formally.

We agree with the reviewer and have added mean calibration and calibration slope to provide an assessment of calibration performance in the Results, Methods, Discussion, and Table 3.

Minor

1. The abstract refers to the 'simulated' TB outcome, but not the observed TB outcome which I believe was also assessed?

We agree it would be ideal to be able to discuss both the observed and simulated outcome in the abstract; however, with only 150 words available, we were unfortunately unable to address this and remain within the word count limit. We present the simulated outcome in the abstract, as this summarizes the most conservative interpretation of model performance. The Results, Discussion, and Conclusion focus on all relevant results, including the observed TB outcome.

2. Lines 98-99 - is this the actual numbers who tested positive for LTBI during the study period? Suggest further clarification of this

Thank you for pointing this out. We have added clarifying text to the manuscript.

3. Line 104 - Is it worth clarifying that linear and spline forms are only relevant to age and BMI? Or are there other variables here that I have missed?

We have clarified this in the text.

4. Consider moving Table S4 (model coefficients) to the main manuscript, since this feels at the "heart" of the paper

We agree with the reviewer's suggestion, and this table has been moved from supplementary materials to main Table 4.

Reviewer #2 (Remarks on code availability):

I did not see a link to the code. Apologies if I missed this

We have included for the code for review in this resubmission, along with a simulated dataset to facilitate understanding of the code functionality.

Reviewer #3 (Remarks to the Author):

I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

Thank you!

REVIEWER COMMENTS

Reviewer #2 (Remarks to the Author):

Many thanks to the authors for their responses. They have done a thorough job of addressing the reviewers' comments. I have 2 further recommendations:

We thank the reviewer for their helpful review. Additional revisions requested below are highlighted in yellow as changes to the manuscript file.

1. Given that the authors acknowledge that this is currently a California-specific tool (given the absence of validation beyond the state), please consider adding "in California" as 2 additional words at the end of the final sentence of the abstract.

We agree with the reviewer and have added this to the abstract.

2. The calibration assessment could still be improved. The authors present "mean calibration". By this, do they mean calibration in the large (<https://www.bmj.com/content/386/bmj-2023-078276>)? If so, the usual ideal target value would be 0, rather than 1. This requires further explanation in the methods section, including how it was calculated and a citation. Alternatively, they could present observed:expected ratios instead of mean calibration. I also recommend adding calibration plots.

Thank you for bringing this to our attention, we apologize for not defining this clearly. Mean calibration is calculated in the same manner as calibration in the large but is then exponentiated, giving it a target value of 1 rather than 0. Exponentiating calibration in the large is suggested in McLernon et al (<https://www.acpjournals.org/doi/10.7326/M22-0844>, the reference cited first in the BMJ article provided by the reviewer for calibration methods), as doing this allows the calibration estimate to approximate the ratio of observed events in the new dataset compared to the number predicted by the risk model. This ratio is closely related to the standardized incidence ratio used in epidemiology. Exponentiating calibration in the large for this purpose is also discussed in the original Crowson et al. manuscript where calibration in the large is derived (<https://journals.sagepub.com/doi/abs/10.1177/0962280213497434>), but doing so is not explicitly defined as mean calibration. We have added further explanation and a citation for McLernon et al to the Methods section to provide the needed clarification for the definition of mean calibration and its relationship to calibration in the large, as suggested by the reviewer.

We agree calibration plots can be very useful tools, however, given the extremely low incidence of TB disease in the population, predictions are highly concentrated on extremely low risks, making these plots hard to read and interpret. As such, we have focused our calibration metrics on mean calibration and calibration slope and added additional focus on real world model performance by evaluating various risk cut-points (reflecting different screening goals) to see how/if new screening choices improve capture of TB disease. We thank the reviewer again for these helpful comments.

Reviewer #3 (Remarks to the Author):

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